

Experiment – 1: Basics of Design Thinking

(a) Explain the Elements of Design (Line, Shape, Color, Texture, Space) with real-life examples.

(Clarity + Examples important)

(b) Identify a real-world problem and propose an innovative solution using design thinking approach.

(a) Elements of Design

The **Elements of Design** are the basic components used to create any visual design. They help in organizing and presenting ideas effectively.

1. Line

A line is a continuous mark that connects two points. It can be straight, curved, thick, or thin.

- **Purpose:** Shows direction, movement, and structure
- **Example:** Road lane markings guide vehicles; notebook lines help in writing neatly

2. Shape

A shape is a closed figure formed by lines, such as circles, squares, and triangles.

- **Purpose:** Helps in identifying objects and organizing layouts
- **Example:** Traffic signs (circle for restrictions, triangle for warnings)

3. Color

Color is used to create visual interest and convey emotions or meanings.

- **Purpose:** Attracts attention and communicates messages
- **Example:** Red indicates danger or stop; green indicates safety or go

4. Texture

Texture refers to the surface quality or feel of an object (smooth, rough, soft, etc.).

- **Purpose:** Adds depth and realism to design
- **Example:** A brick wall has a rough texture, while glass has a smooth texture

5. Space

Space is the area around or between elements in a design. It can be positive (filled) or negative (empty).

- **Purpose:** Improves readability and avoids clutter
- **Example:** Proper spacing in a website or document makes it easy to read

(b) Real-World Problem and Innovative Solution

Problem:

Students often face difficulty in managing their study time and get distracted by mobile phones.

Solution Using Design Thinking Approach:

1. Empathize

Understand that students struggle with focus and time management due to distractions.

2. Define

The main problem is **lack of concentration and poor time management**.

3. Ideate

Develop ideas such as a mobile application that helps students stay focused.

4. Prototype

Create a simple app design with features like:

- Study timer
- Break reminders
- Distraction blocking

5. Test

Test the app with students and collect feedback to improve features.

Final Innovative Solution:

Smart Study Timer App

Features:

- Focus timer (Pomodoro technique)
- Notifications to avoid distractions
- Daily progress tracking

Benefits:

- Improves concentration
- Helps in effective time management
- Reduces mobile distractions

Experiment – 2: Design Principles & Project Idea

(a) Explain the Principles of Design (Balance, Contrast, Emphasis, Unity, Alignment) and discuss their importance in visual communication.

(b) Describe your project idea including:

- Problem statement
- Proposed solution
- Working mechanism

(a) Principles of Design

The **Principles of Design** are guidelines that help in arranging design elements effectively to create visually appealing and meaningful content.

1. Balance

Balance refers to the equal distribution of visual weight in a design.

- **Types:** Symmetrical and Asymmetrical
- **Example:** A website with evenly placed images and text on both sides
- **Purpose:** Creates stability and structure

2. Contrast

Contrast is the difference between elements (color, size, shape) to make them stand out.

- **Example:** Black text on a white background
- **Purpose:** Highlights important information and improves readability

3. Emphasis

Emphasis is used to draw attention to a specific part of the design.

- **Example:** Bold headings or highlighted text
- **Purpose:** Helps users focus on key information

4. Unity

Unity means all elements in a design work together to form a complete and harmonious look.

- **Example:** Using the same color scheme and font style throughout a website
- **Purpose:** Creates consistency and a professional appearance

5. Alignment

Alignment refers to placing elements in a straight line or proper arrangement.

- **Example:** Text aligned to the left or center in a document
- **Purpose:** Makes the design clean and organized

Importance in Visual Communication

- Improves clarity and readability
- Attracts user attention
- Helps convey messages effectively
- Creates a professional and organized layout
- Enhances user experience

(b) Project Idea

Project Title: Smart Attendance System

Problem Statement:

In colleges, attendance is usually taken manually, which is time-consuming and may lead to errors or proxy attendance.

Proposed Solution:

Develop a **Smart Attendance System** using QR code or face recognition technology.

Working Mechanism:

1. **User Login:**
Students and teachers log in to the system
2. **Attendance Marking:**
 - QR Code: Students scan a QR code generated by the teacher
 - OR Face Recognition: System detects and marks attendance automatically
3. **Data Storage:**
Attendance is stored in a database
4. **Report Generation:**
Teachers can view attendance reports anytime

Benefits:

- Saves time
- Reduces manual errors
- Prevents proxy attendance

Experiment – 3: Ideation Techniques

(a) Define Ideation. Explain any two ideation techniques:

- Brainstorming
- Mind Mapping
- SCAMPER (any two)

(b) Apply Design Thinking Stages (Empathize, Define, Ideate, Prototype, Test) to solve a selected problem.

(a) Ideation

Ideation is the process of generating creative ideas and possible solutions to a problem. It is an important stage in Design Thinking where multiple ideas are explored without limitations.

1. Brainstorming

Brainstorming is a technique where a group of people discuss and generate a large number of ideas freely.

- **Key Points:**
 - No criticism of ideas
 - Encourage creative and unusual ideas
 - Focus on quantity over quality

- **Example:**

A team discussing different ideas to reduce traffic problems in a city

2. Mind Mapping

Mind Mapping is a visual technique where ideas are organized around a central concept using branches.

- **Key Points:**
 - Starts with a main idea
 - Related ideas are connected as branches
 - Helps in organizing thoughts clearly

- **Example:**

Creating a mind map for “Online Learning” with branches like advantages, disadvantages, tools, and improvements

(You can write SCAMPER instead of any one above if needed)

(b) Application of Design Thinking Stages

Selected Problem:

Excessive mobile phone usage among students

1. Empathize

Understand users (students) and their behavior.

Students spend too much time on mobile phones, leading to reduced productivity.

2. Define

Clearly state the problem:

“Students are unable to manage their mobile usage, which affects their studies and health.”

3. Ideate

Generate possible solutions:

- Screen time tracking app
- App blocker during study hours
- Reminder notifications

4. Prototype

Design a basic version of the solution:

- Create a simple mobile app interface
- Include features like timer, alerts, and usage tracking

5. Test

Test the app with students:

- Collect feedback
- Identify improvements
- Update features based on user needs

Final Solution:

Screen Time Control App

Features:

- Tracks daily usage
- Sends alerts when usage exceeds limits
- Blocks distracting apps during study time

Experiment – 4: Prototyping & Testing

(a) Explain:

- **Prototyping** (Low fidelity / High fidelity)
- **Testing methods**

Include suitable examples.

(b) Present your solution idea and explain:

- Benefits
- Applications
- Target users

(a) Prototyping & Testing

Prototyping

Prototyping is the process of creating a preliminary version (model) of a product to visualize and test the idea before final development.

Types of Prototyping

1. Low-Fidelity Prototype

- Simple and basic representation of an idea
- Usually created using paper sketches or rough layouts
- Focuses on concept and structure, not details
- **Example:**
Drawing a rough layout of a mobile app on paper

2. High-Fidelity Prototype

- Detailed and interactive version close to the final product
- Includes design elements, colors, and functionality

- **Example:**

A clickable mobile app design created using tools like Figma or Adobe XD

Testing Methods

Testing is done to evaluate the prototype and identify improvements.

1. User Testing

- Real users interact with the prototype
- Feedback is collected based on their experience

2. Feedback Collection

- Asking users questions about usability and design
- Helps in identifying problems and improvements

3. Usability Testing

- Checks how easy and efficient the product is to use
- **Example:**
Students using a study app prototype and suggesting improvements

(b) Solution Idea

Solution Title: Smart Study Planner App

Description:

A mobile application that helps students plan their study schedule, stay focused, and track their progress.

Benefits:

- Improves time management
- Increases productivity
- Reduces distractions
- Helps in achieving academic goals

Applications:

- Schools and colleges
- Competitive exam preparation
- Personal time management

Target Users:

- Students (school and college)
- Competitive exam aspirants
- Anyone who wants to manage time effectively

Experiment – 5: Innovation & Final Solution

- (a) Define **Innovation** and explain how **Design Thinking** helps in **innovation development**.
- (b) Explain your **final solution** with a **neat diagram/flowchart**.

(a) Innovation and Role of Design Thinking

Innovation

Innovation is the process of creating new ideas, products, or improving existing solutions to solve problems more effectively.

- It focuses on **creativity + usefulness**
- It adds value to users and society

👉 **Example:** Smartphones are an innovation that improved communication and daily tasks.

How Design Thinking Helps in Innovation

Design Thinking is a **user-centered approach** that supports innovation by focusing on real user needs.

1. Empathize

- Understand user problems deeply
- Helps in identifying real needs

2. Define

- Clearly state the problem
- Focus on the exact issue

3. Ideate

- Generate multiple creative ideas
- Encourages innovation and thinking out of the box

4. Prototype

- Create a model of the idea
- Makes ideas practical

5. Test

- Get feedback from users
- Improve the solution

Conclusion:

Design Thinking helps in developing innovative solutions that are:

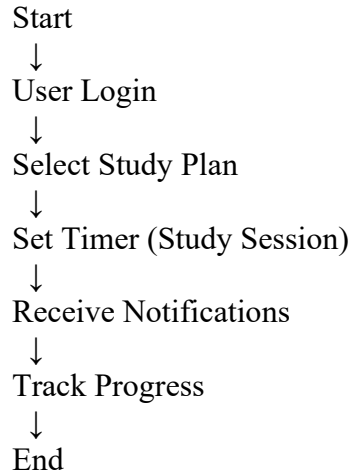
- User-friendly
- Practical
- Effective

(b) Final Solution with Diagram

Final Solution: Smart Study Planner App

Description:

A mobile application that helps students manage their study time, reduce distractions, and improve productivity.

Flowchart:**Explanation:**

- Users log in and choose a study plan
- Timer helps maintain focus
- Notifications remind breaks
- Progress tracking shows performance

Advantages:

- Better time management
- Increased concentration
- Improved academic performance